
The Tangent and Velocity Problems

Work these in class. Worked solutions are on the posted notes page — each problem below names the section heading to look for.

2.1 • Problem 1*see: First: What the Parentheses Mean in $f(x)$*

With $f(x) = x^2 + 1$, write $f(5)$, then $f(a)$, then $f(x + h)$.

2.1 • Problem 2*see: First: What the Parentheses Mean in $f(x)$*

Same function or different? $g(x) = 3x - 4$ and $g(s) = 3s - 4$.

2.1 • Problem 3 (worked together)*see: First, Lines*

The line through $(1, 2)$ with slope 3 — let me do this one both ways.

2.1 • Problem 4*see: First, Lines*

Find the line through $(3, 9)$ with slope 6 — once each way. Do you get the same line?

2.1 • Problem 5 (worked together)*see: Sliding Q Toward P*

One of these first, to set the pattern — the secant from $P = (3, 9)$ to $Q = (6, 36)$.

2.1 • Problem 6*see: Sliding Q Toward P*

Two rows of each table are done. Compute $m_{PQ} = \frac{x^2 - 9}{x - 3}$ for $x = 3.5, 3.1, 3.01$ and for $x = 2.9, 2.99$ — then answer the question that matters: what are the slopes *doing*?

2.1 • Problem 7 (worked together)

see: Evaluating the Limit

Why we are allowed to cancel $(x - 3)$ is the whole point, so let me do this one.

2.1 • Problem 8

see: Average Velocity over Shrinking Intervals

Do not compute another one. Say what number these are heading for, and say how confident you are.

2.1 • Problem 9

see: Why the Two Names Agree

Our intervals all started at $t = 0.5$. What is h if the second time is $t = 0.4$ instead?

2.1 • Problem 10

see: They Are the Same Problem

In the picture, one line's slope is the *average* velocity over $[a, a + h]$ and the other's is the *instantaneous* velocity at $t = a$. Which is which? Then say what happens to the secant as h shrinks toward 0.